

Orthogonalization & QR Decomposition

Linear Algebra & Numerical Methods

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Topics Covered: Orthogonal/Orthonormal Bases, Gram-Schmidt Process, QR Decomposition, Computational Benefits, Modified Gram-Schmidt, Least Squares Applications

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🔗 Learning Objectives

By the end of this week, you will be able to:

- ✔ Understand orthogonal and orthonormal bases and their properties
- ✔ Perform Gram-Schmidt orthogonalization step-by-step
- ✔ Factor any matrix with independent columns as $A = QR$
- ✔ Recognize the computational advantages of orthogonal bases
- ✔ Implement modified Gram-Schmidt for numerical stability
- ✔ Solve least squares problems using QR decomposition
- ✔ Apply QR decomposition in data science applications

💡 Theoretical Foundations

Core Concepts:

- **Orthogonality:** $\mathbf{u}^T \mathbf{v} = 0$
- **Orthonormal Set:** $\mathbf{q}_i^T \mathbf{q}_j = \delta_{ij}$
- **Gram-Schmidt:** Construct orthonormal basis from any independent set
- **QR Theorem:** $A = QR$ where $Q^T Q = I$, R upper triangular
- **Numerical Stability:** Modified Gram-Schmidt prevents error accumulation
- **Least Squares:** $\min_x \|b - Ax\|$ solved via QR

1 Orthogonal and Orthonormal Bases

1.1 Fundamental Definitions

Orthogonal Vectors: Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = 0$$

Orthonormal Set: A set $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k\}$ is orthonormal if:

$$\mathbf{q}_i^T \mathbf{q}_j = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Orthogonal Matrix: A matrix Q whose columns form an orthonormal set:

$$Q^T Q = I \quad (\text{where } I \text{ is the identity matrix})$$

1.2 Key Properties of Orthogonal Matrices

P1. Norm Preservation: $\|Qx\| = \|x\|$ for any vector x

P2. Inverse Simplification: For square Q , $Q^{-1} = Q^T$

P3. Condition Number: $\kappa(Q) = 1$ (optimal numerical stability)

P4. Product Property: Product of orthogonal matrices is orthogonal

P5. Determinant: $|\det(Q)| = 1$ (either +1 or -1)

1.3 Geometric Interpretation

Analogy: An orthogonal matrix represents a rigid rotation or reflection of space. Just as rotating an object doesn't change its shape or size, multiplying by Q preserves vector lengths and angles.

Worked Example

Example: Standard Basis in $\mathbb{R}^3$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

This forms an orthonormal basis. Any vector can be expressed as:

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + v_3\mathbf{e}_3$$

where $v_i = \mathbf{e}_i^T \mathbf{v}$.

2 The Gram-Schmidt Process

2.1 Geometric Motivation

Given linearly independent vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$, we want to construct an orthonormal set $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ spanning the same subspace.

Geometric Idea: At each step, subtract the projection of the current vector onto all previously constructed orthogonal vectors.

2.2 Algorithm

Gram-Schmidt Algorithm

Given: Linearly independent vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$

Step 1:

$$\mathbf{v}_1 = \mathbf{a}_1, \quad \mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}$$

Step j (for $j = 2, \dots, k$):

1. Compute orthogonal component:

$$\mathbf{v}_j = \mathbf{a}_j - \sum_{i=1}^{j-1} (\mathbf{q}_i^T \mathbf{a}_j) \mathbf{q}_i$$

2. Normalize:

$$\mathbf{q}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}$$

2.3 Detailed Example

Worked Example

Example: Gram-Schmidt Process

Let $\mathbf{a}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\mathbf{a}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, $\mathbf{a}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Step 1:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \|\mathbf{v}_1\| = \sqrt{3}, \quad \mathbf{q}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Step 2:

$$\mathbf{q}_1^T \mathbf{a}_2 = \frac{1}{\sqrt{3}}(0 + 1 + 1) = \frac{2}{\sqrt{3}}$$

$$\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix}$$

$$\|\mathbf{v}_2\| = \frac{\sqrt{6}}{3}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

Step 3:

$$\mathbf{q}_1^T \mathbf{a}_3 = \frac{1}{\sqrt{3}}, \quad \mathbf{q}_2^T \mathbf{a}_3 = \frac{1}{\sqrt{6}}$$

$$\mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{6} \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

$$\|\mathbf{v}_3\| = \frac{1}{\sqrt{2}}, \quad \mathbf{q}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

Result: Orthonormal basis:

$$\mathbf{q}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{q}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{q}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

3 QR Decomposition

3.1 The QR Theorem

QR Decomposition Theorem

Any $m \times n$ matrix A with linearly independent columns can be factored as:

$$A = QR$$

where:

- Q is $m \times n$ matrix with orthonormal columns ($Q^T Q = I_n$)
- R is $n \times n$ upper triangular matrix with positive diagonal entries

3.2 Construction from Gram-Schmidt

From Gram-Schmidt, we have:

$$\mathbf{a}_j = \sum_{i=1}^j r_{ij} \mathbf{q}_i \quad \text{where } r_{ij} = \mathbf{q}_i^T \mathbf{a}_j$$

In matrix form:

$$A = [\mathbf{a}_1 \quad \cdots \quad \mathbf{a}_n] = [\mathbf{q}_1 \quad \cdots \quad \mathbf{q}_n] \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ 0 & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_{nn} \end{bmatrix} = QR$$

3.3 Example: QR Decomposition

Worked Example

Example: QR Decomposition

Using previous example with $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

From Gram-Schmidt we have:

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Compute R :

$$r_{11} = \mathbf{q}_1^T \mathbf{a}_1 = \sqrt{3}$$

$$r_{12} = \mathbf{q}_1^T \mathbf{a}_2 = \frac{2}{\sqrt{3}}$$

$$r_{13} = \mathbf{q}_1^T \mathbf{a}_3 = \frac{1}{\sqrt{3}}$$

$$r_{22} = \mathbf{q}_2^T \mathbf{a}_2 = \frac{\sqrt{6}}{3}$$

$$r_{23} = \mathbf{q}_2^T \mathbf{a}_3 = \frac{1}{\sqrt{6}}$$

$$r_{33} = \mathbf{q}_3^T \mathbf{a}_3 = \frac{1}{\sqrt{2}}$$

Thus:

$$R = \begin{bmatrix} \sqrt{3} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{\sqrt{6}}{3} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Verify: $QR = A$.

4 Vector Projections and Orthogonal Decomposition

4.1 Projection onto a Vector

The projection of vector $\mathbf{v}$ onto vector $\mathbf{u}$ is:

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{u}^T \mathbf{v}}{\|\mathbf{u}\|^2} \mathbf{u}$$

The orthogonal component is:

$$\mathbf{v}_{\perp} = \mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v})$$

4.2 Projection onto a Subspace

For a subspace S with orthonormal basis $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$, the projection of $\mathbf{x}$ onto S is:

$$\text{proj}_S(\mathbf{x}) = \sum_{i=1}^k (\mathbf{q}_i^T \mathbf{x}) \mathbf{q}_i = Q Q^T \mathbf{x}$$

where Q has columns $q_1, \dots, q_k$.

4.3 Orthogonal Decomposition Theorem

Every vector $\mathbf{x} \in \mathbb{R}^n$ can be uniquely written as:

$$\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$$

where:

- $\mathbf{x}_1 \in S$ (projection onto subspace S)
- $\mathbf{x}_2 \in S^\perp$ (orthogonal to S)
- $\mathbf{x}_1^T \mathbf{x}_2 = 0$

5 Least Squares Problems and QR Decomposition

5.1 Geometric Interpretation

The least squares problem $\min_x \|b - Ax\|$ seeks the vector $\hat{b}$ in the column space of A that is closest to b . This is exactly the projection of b onto $\text{Col}(A)$.

5.2 Solution via QR Decomposition

Given $A = QR$, the normal equations $A^T Ax = A^T b$ become:

$$R^T Q^T Q R x = R^T Q^T b \quad \Rightarrow \quad R^T R x = R^T Q^T b$$

Since R is invertible, this simplifies to:

$$R x = Q^T b$$

which is a triangular system that can be solved by back substitution.

5.3 Step-by-Step Example

Worked Example

Example: Least Squares via QR

Solve $Ax \approx b$ where:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Step 1: QR Decomposition (using first two columns from earlier):

$$Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{3} & \frac{2}{\sqrt{3}} \\ 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$$

Step 2: Compute $Q^T b$:

$$Q^T b = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{6}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} 2\sqrt{3} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$$

Step 3: Solve $Rx = Q^T b$:

$$\begin{bmatrix} \sqrt{3} & \frac{2}{\sqrt{3}} \\ 0 & \frac{\sqrt{6}}{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2\sqrt{3} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$$

From second equation: $\frac{\sqrt{6}}{3}x_2 = \frac{2}{\sqrt{6}} \Rightarrow x_2 = 1$

From first equation: $\sqrt{3}x_1 + \frac{2}{\sqrt{3}} \cdot 1 = 2\sqrt{3} \Rightarrow x_1 = \frac{4}{3}$

Solution: $x = \begin{pmatrix} \frac{4}{3} \\ 1 \end{pmatrix}$

6 Modified Gram-Schmidt for Numerical Stability

6.1 Problem with Classical Gram-Schmidt

In finite-precision arithmetic, classical Gram-Schmidt can lose orthogonality due to:

- Accumulation of rounding errors
- Cancellation in subtraction of nearly equal vectors
- Using original vectors rather than updated ones

6.2 Modified Algorithm

Modified Gram-Schmidt Algorithm

Given: $\mathbf{a}_1, \dots, \mathbf{a}_n$

1. Set $\mathbf{v}_1 = \mathbf{a}_1$, then $\mathbf{q}_1 = \mathbf{v}_1 / \|\mathbf{v}_1\|$

2. For $j = 2, \dots, n$:

(a) Set $\mathbf{v} = \mathbf{a}_j$

(b) For $i = 1, \dots, j - 1$:

$$\mathbf{v} = \mathbf{v} - (\mathbf{q}_i^T \mathbf{v}) \mathbf{q}_i$$

(c) Set $\mathbf{v}_j = \mathbf{v}$

(d) $\mathbf{q}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$

6.3 Comparison

Classical Gram-Schmidt

Uses original $\mathbf{a}_j$ for all projections

Can lose orthogonality due to rounding errors

Easier to understand mathematically
Same result in exact arithmetic

Modified Gram-Schmidt

Uses updated $\mathbf{v}$ for subsequent projections

More numerically stable

Better for computer implementation
Same result in exact arithmetic

Practical Applications in Data Science & ML

Practical Applications in Data Science & ML:

- **Principal Component Analysis (PCA):** QR decomposition used in eigenvalue computation
- **Least Squares Regression:** Stable solution of normal equations
- **Recommender Systems:** Matrix factorization techniques
- **Image Compression:** Via orthogonal transforms (DCT in JPEG)
- **Signal Processing:** Filter design and noise reduction
- **Computer Vision:** Camera calibration and 3D reconstruction
- **Natural Language Processing:** Word embedding orthogonalization
- **Quantum Computing:** Quantum state representation
- **Robotics:** Coordinate transformations
- **Control Systems:** State estimation and filtering

7 Implementation in MATLAB and Python

7.1 MATLAB Implementation

```
% QR Decomposition  
A = [1 0 0; 1 1 0; 1 1 1];
```

```

[Q, R] = qr(A); % Full QR decomposition
[Q1, R1] = qr(A, 0); % Economy-size QR

% Orthonormal basis for column space
Q_orth = orth(A);

% Solving least squares via QR
x = A \ b; % Uses QR internally

% Gram-Schmidt implementation
function [Q, R] = classicalGS(A)
    [m, n] = size(A);
    Q = zeros(m, n);
    R = zeros(n, n);
    for j = 1:n
        v = A(:, j);
        for i = 1:j-1
            R(i, j) = Q(:, i)' * A(:, j);
            v = v - R(i, j) * Q(:, i);
        end
        R(j, j) = norm(v);
        Q(:, j) = v / R(j, j);
    end
end
end

```

7.2 Python Implementation

```

import numpy as np
from scipy.linalg import qr, orth

# QR Decomposition
A = np.array([[1, 0, 0], [1, 1, 0], [1, 1, 1]])
Q, R = qr(A, mode='full') # Full QR
Q1, R1 = qr(A, mode='economic') # Economy-size

# Orthonormal basis for column space
Q_orth = orth(A)

# Solving least squares via QR
x = np.linalg.lstsq(A, b, rcond=None)[0]

# Gram-Schmidt implementation
def modified_gram_schmidt(A):
    m, n = A.shape
    Q = np.zeros((m, n))
    R = np.zeros((n, n))

    for j in range(n):
        v = A[:, j].copy()
        for i in range(j):
            R[i, j] = np.dot(Q[:, i], v)
            v = v - R[i, j] * Q[:, i]

```

```

R[j, j] = np.linalg.norm(v)
Q[:, j] = v / R[j, j]

return Q, R

```

8 Computational Considerations

8.1 Complexity Analysis

- **Gram-Schmidt:** $O(mn^2)$ operations for $m \times n$ matrix
- **Householder QR:** $O(mn^2)$ but more stable than Gram-Schmidt
- **Givens Rotations:** $O(mn^2)$ but good for sparse matrices
- **Storage:** Q can be stored implicitly in Householder form

8.2 Numerical Considerations

1. **Condition Number:** Orthogonal matrices have $\kappa(Q) = 1$
2. **Stability:** Modified Gram-Schmidt or Householder preferred
3. **Reorthogonalization:** May be needed for ill-conditioned matrices
4. **Pivoting:** Column pivoting QR: $AP = QR$

Learning Resources

Recommended Resources:

- **Textbook:** Gilbert Strang, "Linear Algebra and Its Applications"
- **Numerical:** Golub & Van Loan, "Matrix Computations"
- **Online Course:** MIT 18.06 Linear Algebra (OpenCourseWare)
- **Software:** MATLAB/Octave: `qr()` function
- **Python:** `numpy.linalg.qr()`, `scipy.linalg.qr()`
- **Visualization:** 3Blue1Brown Linear Algebra series (YouTube)

Summary

- Orthogonal/orthonormal bases simplify many linear algebra computations
- Gram-Schmidt converts any basis to an orthonormal one
- QR decomposition $A = QR$ factors any matrix with independent columns
- Modified Gram-Schmidt improves numerical stability
- QR decomposition is fundamental for least squares and eigenvalue problems
- Orthogonal transformations preserve norms and condition numbers
- Applications span data science, machine learning, signal processing, and more

Practice Problems

Practice Problems

Problem 1: Basic Orthogonality

1. Show that vectors $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$ are orthogonal.
2. Normalize these vectors to create an orthonormal set.
3. Find a third vector orthogonal to both.

Problem 2: Gram-Schmidt Process Apply Gram-Schmidt to:

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{a}_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Problem 3: QR Decomposition Find QR decomposition of:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Problem 4: Least Squares with QR Use QR decomposition to solve:

$$\min_x \left\| \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} \right\|$$

Problem 5: Application in PCA Given data points: (1, 2), (2, 4), (3, 3), (4, 5)

1. Compute the first principal component
2. Project the data onto this component
3. Reconstruct the data from the projection

Solutions to Selected Problems

Problem 1 Solution:

$$1. \mathbf{u}^T \mathbf{v} = 1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 0 = 2 - 2 + 0 = 0$$

$$2. \|\mathbf{u}\| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}, \text{ so } \hat{\mathbf{u}} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\|\mathbf{v}\| = \sqrt{2^2 + (-1)^2 + 0^2} = \sqrt{5}, \text{ so } \hat{\mathbf{v}} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$$

$$3. \text{ For } \mathbf{w} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ orthogonal to both:}$$

$$\mathbf{u}^T \mathbf{w} = x + 2y - z = 0$$

$$\mathbf{v}^T \mathbf{w} = 2x - y = 0 \Rightarrow y = 2x$$

$$\text{Substitute: } x + 2(2x) - z = 0 \Rightarrow 5x - z = 0 \Rightarrow z = 5x$$

$$\text{So } \mathbf{w} = x \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}. \text{ For unit length: } \mathbf{w} = \frac{1}{\sqrt{30}} \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$$

$$\textbf{Problem 4 Solution: Let } A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

$$\text{Apply Gram-Schmidt: } \mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \|\mathbf{a}_1\| = \sqrt{3}, \mathbf{q}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\mathbf{a}_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \mathbf{q}_1^T \mathbf{a}_2 = \frac{6}{\sqrt{3}} = 2\sqrt{3} \quad \mathbf{v}_2 = \mathbf{a}_2 - 2\sqrt{3}\mathbf{q}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \|\mathbf{v}_2\| = \sqrt{2},$$

$$\mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{So } Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \end{bmatrix}, R = \begin{bmatrix} \sqrt{3} & 2\sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}$$

$$Q^T \mathbf{b} = \begin{bmatrix} \frac{10}{\sqrt{3}} \\ \frac{3}{\sqrt{2}} \end{bmatrix}$$

$$\text{Solve } R\mathbf{x} = Q^T \mathbf{b}: \sqrt{2}x_2 = \frac{3}{\sqrt{2}} \Rightarrow x_2 = \frac{3}{2} \quad \sqrt{3}x_1 + 2\sqrt{3} \cdot \frac{3}{2} = \frac{10}{\sqrt{3}} \Rightarrow x_1 = \frac{1}{3}$$

$$\text{Solution: } \mathbf{x} = \begin{pmatrix} \frac{1}{3} \\ \frac{3}{2} \end{pmatrix}$$

Next Week Preview

Next Week Preview: Eigenvalues and Eigenvectors

- Definition: $A\mathbf{x} = \lambda\mathbf{x}$
- Characteristic polynomial and equation
- Diagonalization of matrices
- Spectral theorem for symmetric matrices
- Singular Value Decomposition (SVD)
- Applications to data science and ML